Spring 2026 Graph Theory Project Assignment

All students will have a choice to do one of the following projects. You may work in teams of up to two students. Students working in a team of two will be held to a more stringent standard than students working alone.

1) Assemble a PowerPoint-style presentation with at least 15 slides (not counting title or references) that could be used in a presentation about a topic of your choice. You will not give the presentation, but you will submit your slides as your final project.

   Popular topics include applications of various families of graphs (fancy sorts of trees are *very* popular); vertex- or edge-coloring algorithms or heuristics; graph drawing algorithms; graph-based efficient storing, searching, or sorting techniques; network flows (max-flow min-cut problems); tournament ranking schemes.

   I would suggest the following framework:

   a) Outline of the presentation
   b) Announce the topic and introduce notation as needed
   c) Provide some historical context
   d) Develop the theoretical framework behind the topic
   e) Current state of the topic and where it seems to be headed
   f) References

2) Alternatively, students may choose to construct a "graph theorist's sketchpad" application that mimics but expands on the sketcher I wrote and use for in-class examples.

3) A third option is for students to construct a "WSU-GPS" where the application overlays a graph structure on a map of WSU (vertices are campus locations and edges would represent direct walking routes between locations.) One would then implement the appropriate graph traversal algorithms to suggest optimal routes.

4) If you want a numerical project, you can build an application that accepts graph data and returns linear algebraic information—this can concern the incidence matrix, adjacency matrix A, Laplacian matrix L, etc., including eigenstructures (of A or L) –and interpretations of what this information reveals about the graph. You are allowed to use canned eigenvalue/eigenvector routines if you go this route.

5) Another choice of appropriate application project would be based on random graphs (e.g. random walks on graphs or sequences of graphs generated randomly, uniform sampling of graphs).

a. One option for this is to code an MCMC model like the one in this paper for decrypting substitution cyphers: https://probability.ca/jeff/ftpdir/decipherart.pdf I will provide a cypher text for you to try to decrypt.
b. Another option is to code the MCMC model in this paper: https://arxiv.org/pdf/2411.17613. It provides a method for uniform sampling of trees (both labeled and unlabeled). You can use this to get an estimate of the number of unlabeled trees up to isomorphism, then compare your results to the OEIS entry: https://oeis.org/A000081

6) Adapting ideas from the "Popular topics" listed in Project (1) for a software application would also be a worthy project idea.

If you take a software application approach, I will expect you to demonstrate it via screen sharing during office hours toward the end of the semester or during the final exam week for this course. Also, you must provide a short (3 to 5 page) narrative of your application's features and how you implemented your routines.

The final document submissions are due at 5:00 on Monday, May 4th. Demonstrations of coding applications during finals week will take place via Zoom at a time to be determined.